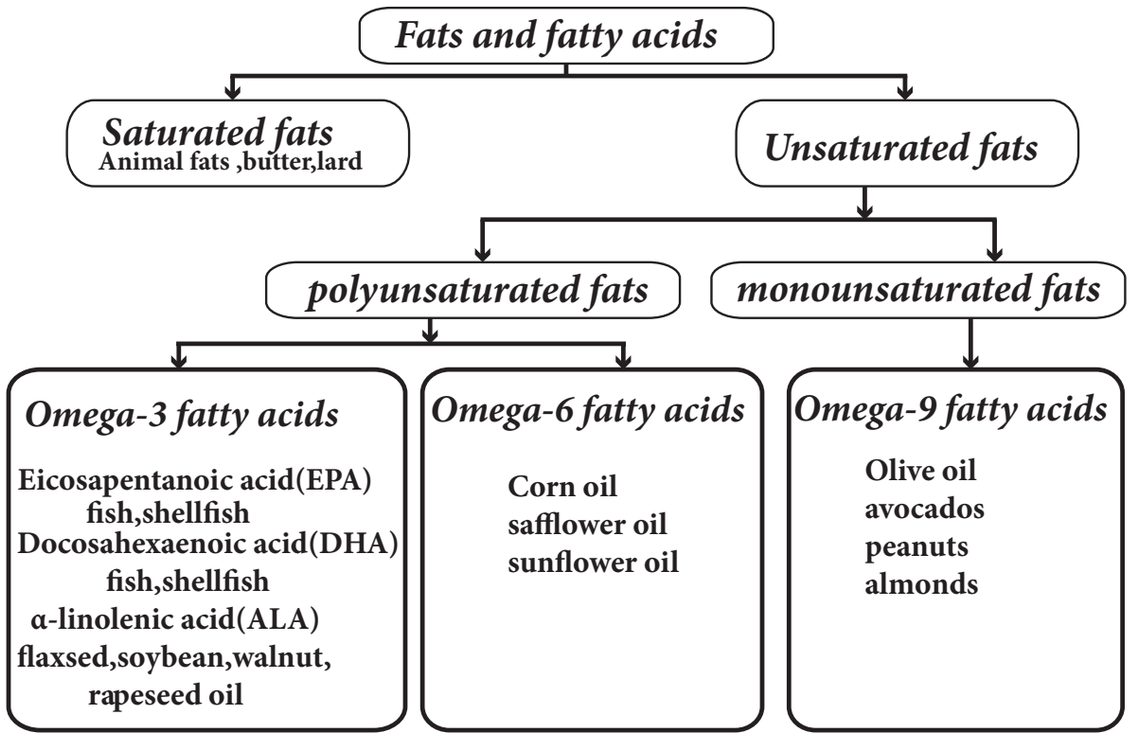


Fig 10.15: Classification of fats based on fatty acids

weight loss. Foods which contain MUFA (Oleic acid) are avocados, olives, olive oil, peanut butter and peanut oil. It is also known as omega-9 fatty acid.

(ii **Polyunsaturated fatty acid (PUFA):** A fatty acid is polyunsaturated if it contains more than one double bond. They are of two types, namely Omega-3 and Omega-6 fatty acids.

a) **Omega-3:** It is also called ω−3 fatty acids or n−3 fatty acids with a double bond (C=C) at the third carbon atom from the end of the carbon chain. The three types of omega−3 fatty acids involved in human physiology are α-linolenic acid (ALA) [found in plant oils], eicosapentaenoic acid (EPA), and docosahexaenoic acid (DHA) [both commonly found in marine oils]. Common sources of plant oils containing the omega−3 ALA fatty acid include walnut, flaxseed, flaxseed oil,soybeans and chia seeds.The sources of animal omega−3 EPA and DHA fatty acids include fish and fish oils.

The health benefits of omega-3 fatty acids are immense and they have been proven effective in the treatment and prevention of hundreds of medical conditions which includes high cholesterol, depression, anxiety, cancer, diabetes mellitus, inflammatory diseases, arthritis and cardiovascular diseases.

b) **Omega-6:** Omega-6 fatty acids (also referred to as ω-6 fatty acids or n-6 fatty acids) are a family of pro-inflammatory and anti-inflammatory polyunsaturated fatty acids that have in common a final carbon-carbon double bond in the n-6 position, that is the sixth bond, counting from

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